How to Read This Cover Letter:

1. The purpose of this document is to convince you that I indeed understand QEC codes.

2. If the math is boring, just read the bold text.

Disclaimer: I wrote this draft myself. AI was used to check the grammar and help insert and format the mathematical equations. I reviewed every line, and the mathematical content and final wording are my responsibility.

1 I Have a Chain-Complex-Native Brain

I spent years studying homological algebra and category theory. So when I came across Kitaev's toric code, the X- and Z-checks immediately looked like boundary and coboundary maps to me.

$$0 \longrightarrow \text{plaquettes } (Z\text{-checks}) \xrightarrow{\partial_2} \text{edges (physical qubits)} \xrightarrow{\partial_1} \text{vertices } (X\text{-checks}) \longrightarrow 0.$$

1.1 For the Math friend to read

One useful approach to constructing qLDPC codes and CSS code is through chain complexes.

Let

$$C_\bullet : \quad C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

be a three-term chain complex of finite-dimensional vector spaces over $\mathbb{F}_2$, with

$$\partial_1 \circ \partial_2 = 0.$$

After choosing bases, we obtain a CSS quantum code by defining its parity-check matrices as

$$\boxed{H_X = \partial_1, \qquad H_Z = \partial_2^{\mathsf{T}}.}$$

The chain-complex condition becomes

$$\boxed{H_X H_Z^{\mathsf{T}} = \partial_1 \partial_2 = 0,}$$

which is precisely the commutativity condition for the X- and Z-type stabilizers.

In this language, the physical qubits correspond to basis elements of C_1, while the X- and Z-type stabilizer generators correspond to basis elements of C_0 and C_2, respectively.

The number of physical qubits is

$$n = \dim_{\mathbb{F}_2} C_1.$$

More importantly, the number of encoded logical qubits is determined by the first homology group:

$$\boxed{H_1(C_\bullet) = \frac{\ker \partial_1}{\operatorname{im} \partial_2}, \qquad k_{\text{logical}} = \dim H_1(C_\bullet).}$$

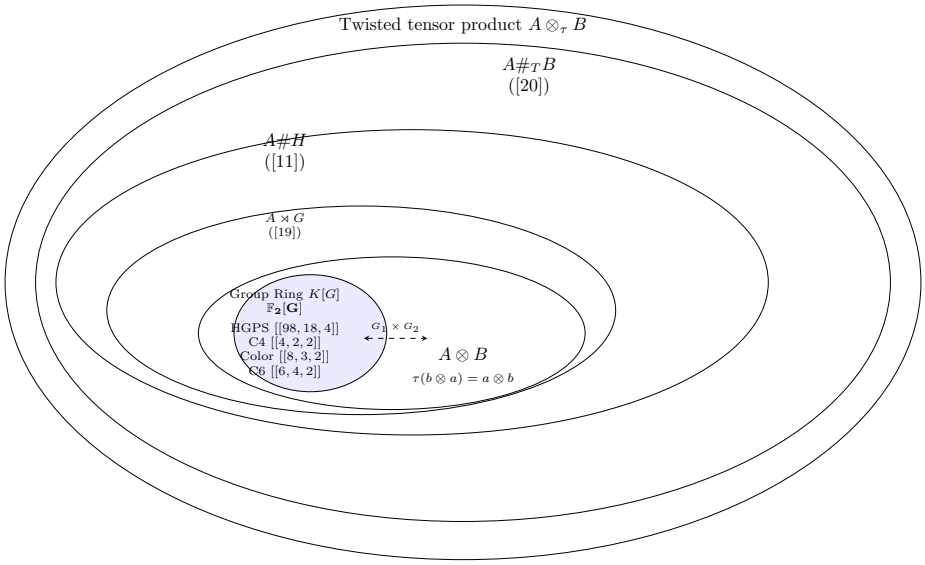
Thus, logical operators are naturally described by cycles that are not boundaries, or cocycles that are not coboundaries.This motivates a homological formulation of the code-construction problem.

For me personally, this formalization made understanding and picking up QEC feel like running into your own mother in a foreign country.

2 My research training has made me looking at all the current QEC from a even more generalized angle

In quantum error correction (QEC), taking tensor products is a way to generate new codes.

My Ph.D. research focused on a **generalization** of the tensor product called the **twisted tensor product**, which encodes more complex "gluing conditions." There relationship is viewed as follow: (detailed definition of each ring see table 1 at section 5.



And just like we can take the twisted tensor product of two algebras A and B and get $A \otimes_\tau B$, we can study twisted tensor products of differential graded algebras(i.e. chains) too, and get something like

$$\boxed{C_\bullet \otimes_{\widetilde{\tau}} D_\bullet,}$$

where $\widetilde{\tau}$ is a twisting map compatible with the differentials.

My reseach devoted my Ph.D. to studying in what situations the Ext-algebra of a twisted tensor product is again a twisted tensor product. I studied the homological algebra of twisted tensor products. **So heuristically, think of $E(A)$ as homological information attached to A, but with an algebra structure. And I studied under what specific conditions we would have**

$$\boxed{E(A \otimes_\tau B) \cong E(B) \otimes_{\tau^!} E(A).}$$

In particular, my theorem considers Koszul algebras A and B with a strongly graded, invertible twisting map τ, under the hypotheses of my dissertation. Here $\tau^!$ is the induced twisting map on their Ext-algebras. **All the popular codes on the market can be easily understood through this framework.**

3 Examples of Error Correction Codes

The following section is partly to show off that I do know QEC codes, and partly to give the reader an idea of how I understand them! You can skip them all, all the condensed information are in section table and charts. 5. I grouped the codes by its underlying algebra, one sections for bland tensor products 3.1, and one section for twisted tensor products with non-trivial twist 3.2.

3.1 Examples of Codes as tensor product

Iceberg code $[[4, 2, 2]]$**.** Let

$$A = \frac{\mathbb{F}_2[x]}{(x^2 - 1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^2 - 1)}.$$

Then

$$R = A \otimes_{\tau_{\text{flip}}} B,$$

where

$$\boxed{\tau_{\text{flip}}(y^j \otimes x^i) = x^i \otimes y^j.}$$

So this is just the ordinary tensor product.

Let

$$u = 1 + x, \qquad v = 1 + y, \qquad I = (u, v) \subseteq R.$$

Now we have a short exact sequence of $\mathbb{F}_2$-vector spaces $C_\bullet$

$$\boxed{0 \longrightarrow \mathbb{F}_2 \xrightarrow{\iota_A} A \xrightarrow{\varepsilon_A} \mathbb{F}_2 \longrightarrow 0,}$$

where the maps are explicitly given by

$$\boxed{\begin{aligned} \iota_A(1) &= 1 + x = u, \\ \varepsilon_A(a_0 + a_1 x) &= a_0 + a_1. \end{aligned}}$$

And similarly, for B, $D_\bullet$

$$\boxed{0 \longrightarrow \mathbb{F}_2 \xrightarrow{\iota_B} B \xrightarrow{\varepsilon_B} \mathbb{F}_2 \longrightarrow 0,}$$

with

$$\boxed{\begin{aligned} \iota_B(1) &= 1 + y = v, \\ \varepsilon_B(b_0 + b_1 y) &= b_0 + b_1. \end{aligned}}$$

Notice that

$$\varepsilon_A \circ \iota_A = 0, \qquad \varepsilon_B \circ \iota_B = 0.$$

In fact, both sequences are exact.

Now, by the Künneth formula, the ordinary tensor product of these two chain complexes has

$$\boxed{H_n(C_\bullet \otimes_{\mathbb{F}_2} D_\bullet) \cong \bigoplus_{p+q=n} H_p(C_\bullet) \otimes_{\mathbb{F}_2} H_q(D_\bullet) = 0.}$$

So the ordinary tensor-product complex is exact too.

But for the Iceberg code, we can take the degreewise tensor product of the two sequences instead.

That gives the chain complex

$$\boxed{0 \longrightarrow \mathbb{F}_2 \otimes \mathbb{F}_2 \xrightarrow{\partial_2} A \otimes B \xrightarrow{\partial_1} \mathbb{F}_2 \otimes \mathbb{F}_2 \longrightarrow 0,}$$

where

$$\boxed{\partial_2 = \iota_A \otimes \iota_B, \qquad \partial_1 = \varepsilon_A \otimes \varepsilon_B.}$$

The differentials are explicitly

$$\boxed{\begin{aligned} \partial_2(1) &= (1 + x) \otimes (1 + y) \\ &= 1 + x + y + xy, \end{aligned}}$$

and

$$\boxed{\partial_1(x^i y^j) = 1, \qquad i, j \in \{0, 1\}.}$$

Here we identify $\mathbb{F}_2 \otimes \mathbb{F}_2$ with $\mathbb{F}_2$. Thus,

$$\boxed{0 \longrightarrow \mathbb{F}_2 \xrightarrow{\cdot(1+x)(1+y)} R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0.}$$

Here the first arrow sends 1 to $(1+x)(1+y)$.

And because

$$\varepsilon((1+x)(1+y)) = 0,$$

over characteristic 2, we have

$$\boxed{\partial_1 \partial_2 = 0.}$$

This concludes the parity check !

Moreover, because the kernel of $\partial_1 = \varepsilon$ is the ideal generated by $1+x$ and $1+y$,

$$\boxed{\ker \partial_1 = I = (1+x, 1+y),}$$

and

$$I^2 = \mathbb{F}_2 \cdot (1+x)(1+y),$$

we have

$$\boxed{\mathbb{F}_2 \cong I^2 \quad \text{as } \mathbb{F}_2\text{-modules.}}$$

$$\boxed{\begin{array}{l}\phi : \mathbb{F}_2 \xrightarrow{\sim} I^2, \\ \qquad 1 \longmapsto (1+x)(1+y).\end{array}}$$

We thus have an isomorphism of chain complexes

$$\boxed{\left(0 \longrightarrow \mathbb{F}_2 \xrightarrow{\partial_2} R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0\right) \cong \left(0 \longrightarrow I^2 \xhookrightarrow{\iota} R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0\right)}$$

in the category Ch(R-Mod). Thus the two chain complexes are isomorphic, not merely quasi-isomorphic. Consequently, their first homology groups are isomorphic.

This establishes the chain-complex formulation of the $[[4,2,2]]$ Iceberg code.

C6 code $[[6,4,2]]$**.** Take

$$A = \frac{\mathbb{F}_2[x]}{(x^2-1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^3-1)}.$$

Then

$$R = A \otimes B$$

.

The algebra R is also isomorphic to $\mathbb{F}_2[z]/(z^6-1)$.

Let

$$N = 1 + z + z^2 + z^3 + z^4 + z^5.$$

The complex

$$0 \longrightarrow \langle N \rangle \hookrightarrow R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0$$

gives the $[[6,4,2]]$ code.

Eight-qubit CSS color code $[[8,3,2]]$**.** Take

$$A = \frac{\mathbb{F}_2[x]}{(x^2-1)},$$

$$B = \frac{\mathbb{F}_2[y,z]}{(y^2-1, z^2-1)}.$$

Then

$$R = A \otimes_\tau B,$$

Let

$$I = (1+x, 1+y, 1+z) \subseteq R.$$

The augmentation-ideal complex

$$0 \longrightarrow I^2 \hookrightarrow R \xrightarrow{\varepsilon} \mathbb{F}_2 \longrightarrow 0$$

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Many-hypercube $[[16,4,4]]$**.** Now let's consider the $[[16,4,4]]$ many-hypercube code, which is related to the $[[4,2,2]]$ Iceberg code through concatenation.

Again, we start with two algebras whose generators have finite order:

$$A = \frac{\mathbb{F}_2[x,y]}{(x^2-1, y^2-1)},$$

$$B = \frac{\mathbb{F}_2[p,q]}{(p^2-1, q^2-1)}.$$

We take their ordinary tensor product

$$\boxed{R = A \otimes B,}$$

Specifically,

$$\boxed{R \cong \frac{\mathbb{F}_2[x,y,p,q]}{(x^2-1, y^2-1, p^2-1, q^2-1)}.}$$

Now let

$$u = 1+x, \qquad v = 1+y,$$

and define two elements of B:

$$P = 1+pq, \qquad Q = p+q.$$

Define the ideal

$$\boxed{J = (uv,\ uP+vQ,\ vP+uQ) \subseteq R.}$$

We can write J explicitly as an $\mathbb{F}_2$-vector space:

$$\boxed{\begin{aligned} J = \operatorname{span}_{\mathbb{F}_2} \big\{ & uv,\ puv,\ quv,\ pquv, \\ & uP + vQ,\ vP + uQ \big\}. \end{aligned}}$$

Furthermore, using $u^2 = v^2 = P^2 = Q^2 = PQ = 0$, we obtain

$$\boxed{J^2 = 0.}$$

This is exactly the algebraic condition we need to construct our chain complex.

The chain complex.

Define the linear functional

$$\lambda : R \longrightarrow \mathbb{F}_2$$

by chopping off all nonconstant terms and keeping only the constant term in the polynomial .

For any $r, s \in R$, let

$$\langle r, s \rangle = \lambda(rs).$$

Now consider the chain complex

$$\boxed{0 \longrightarrow J \xhookrightarrow{\partial_2} R \xrightarrow{\partial_1} J^* \longrightarrow 0,}$$

where

$$J^* = \operatorname{Hom}_{\mathbb{F}_2}(J, \mathbb{F}_2).$$

The first differential is inclusion:

$$\boxed{\partial_2(j) = j.}$$

The second differential is explicitly

$$\boxed{\begin{aligned} & \partial_1 : R \longrightarrow J^*, \\ & \qquad r \longmapsto \big(j \longmapsto \lambda(rj) \big). \end{aligned}}$$

Because $J^2 = 0$, for any $j, j' \in J$,

$$\partial_1(\partial_2(j))(j') = \lambda(jj') = 0.$$

Thus,

$$\boxed{\partial_1 \partial_2 = 0.}$$

We have constructed the many-hypercube code starting from the tensor product of two algebras and its generating relations to obtain

$$\boxed{0 \longrightarrow J \longrightarrow R \longrightarrow J^* \longrightarrow 0.}$$

Its logical qubits are described by

$$\boxed{H_1 \cong \frac{\mathrm{Ann}_R(J)}{J}.}$$

Knill's C6 code $[[6, 2, 2]]$**.** Let

$$A = \frac{\mathbb{F}_2[x]}{(x^2 - 1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^3 - 1)}.$$

Then

$$R = A \otimes B.$$

Define the ideal

$$\boxed{J = ((1 + x)(1 + y)) \subseteq R.}$$

Now define

$$\boxed{0 \longrightarrow J \xhookrightarrow{\partial_2} R \xrightarrow{\partial_1} J^* \longrightarrow 0,}$$

where

$$\partial_1(r)(j) = \lambda(rj^*).$$

$$\partial_2(j) = j$$

. Then

$$\partial_1(r)(j) = \lambda(rj^*).$$

Since J is stable under $*$ and $J^2 = 0$, we obtain

$$\boxed{\partial_1\partial_2 = 0.}$$

giving the code

$$\boxed{[[6, 2, 2]].}$$

3.2 Examples of Codes with Nontrivial Twisted Tensor Product Structures

Non-Abelian two-block group-algebra (2BGA) codes. Another example is the non-Abelian 2BGA code. Let

$$A = \frac{\mathbb{F}_2[x]}{(x^3 - 1)}, \qquad B = \frac{\mathbb{F}_2[y]}{(y^2 - 1)}.$$

This time, we take a nontrivial twisted tensor product

$$\boxed{R = A \otimes_\tau B,}$$

where

$$\boxed{\tau(y^j \otimes x^i) = x^{(-1)^j i} \otimes y^j.}$$

So $yx = x^{-1}y$, and the algebra is noncommutative. For $a, b \in R$, define

$$\boxed{H_X = [L(a) \mid R(b)], \qquad H_Z = [R(b)^T \mid L(a)^T].}$$

Here $L(a)$ and $R(b)$ denote left and right multiplication, respectively. Because left and right multiplication commute, we have $H_X H_Z^T = 0$. We can actually write out the chain complex:

$$\boxed{0 \longrightarrow R \xrightarrow{\begin{bmatrix} R(b) \\ L(a) \end{bmatrix}} R^2 \xrightarrow{\begin{bmatrix} L(a) & R(b) \end{bmatrix}} R \longrightarrow 0.}$$

Moreover, this twist satisfies the compatibility conditions in Definition 1.

Eight-qubit Gottesman code $[[8, 3, 3]]$**.** Another interesting example is the eight-qubit Gottesman code, which is non-CSS.

Over $\mathbb{C}$, let

$$A = \mathbb{C}[C_2^8], \qquad B = \mathbb{C}[C_2^8],$$

where A represents the X operators and B represents the Z operators. The underlying Pauli operator algebra is indeed a twisted tensor product

$$\boxed{P = A \otimes_\tau B,}$$

where the twisting map is

$$\boxed{\tau(z^{\mathbf{v}} \otimes x^{\mathbf{u}}) = (-1)^{\mathbf{u}\cdot\mathbf{v}} x^{\mathbf{u}} \otimes z^{\mathbf{v}}.}$$

In particular,

$$\boxed{z_i x_i = -x_i z_i.}$$

This is a genuinely nontrivial twisting map, satisfying the unit and compatibility conditions in Definition 1.

With special choice of H and J, we can still associate a chain complex to the code:

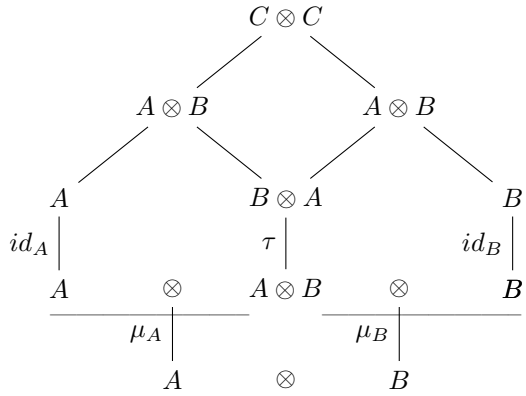
$$\boxed{0 \longrightarrow \mathbb{F}_2^5 \xrightarrow{H^T} \mathbb{F}_2^{16} \xrightarrow{HJ} \mathbb{F}_2^5 \longrightarrow 0.}$$

4 What is a twisted tensor product

During my Ph.D., I studied a construction in noncommutative algebra called the **twisted tensor product**, denoted $A \otimes_\tau B$, where τ is a map from $B \otimes A$ to $A \otimes B$:

$$\tau : B \otimes A \longrightarrow A \otimes B.$$

It generalizes the ordinary tensor product. If we denote $C = A \otimes_\tau B$, its multiplication looks like the diagram below, where the diagram is read from top to bottom.



For a regular tensor, the τ will be the swap map between the two vector space A and B.

The formal definition of such a twist is as follows.

Definition 1. Let (A, μ_A) and (B, μ_B) be k-algebras, where μ_A and μ_B denotes the multiplications on A and B respectivly. A k-linear map

$$\tau : B \otimes A \longrightarrow A \otimes B$$

is an **algebra twist** if and only if $\tau(b \otimes 1) = 1 \otimes b$, $\tau(1 \otimes a) = a \otimes 1$, and the following two compatibility condition is satisfied:

$$(1_A \circ \mu_B) \circ (\tau \otimes 1_A) \circ (1_B \otimes \tau) = \tau \circ (\mu_B \otimes 1_A) \tag{1}$$

$$(\mu_A \circ 1_B) \circ (1_A \otimes \tau) \circ (\tau \otimes 1_A) = \tau \circ (1_B \otimes \mu_A) \tag{2}$$

5 Tables and Graphs

Construction / code	**Multiplication**	**Twisting map** $\tau\colon B\otimes A\to A\otimes B$
I. Algebra constructions		
Twisted tensor product $A\otimes_\tau B$	If $\tau(b\otimes a')=\sum_i a_i'\otimes b_i$, then $(a\otimes b)(a'\otimes b')=\sum_i aa_i'\otimes b_ib'$.	$\tau(b\otimes a)=\sum_i a_i\otimes b_i$, subject to the twisting identities.
Ordinary tensor product $A\otimes B$	$(a\otimes b)(a'\otimes b')=aa'\otimes bb'$.	$\tau(b\otimes a)=a\otimes b$ (the flip).
Hopf smash product $A\#H$	For $\Delta(h)=\sum h_{(1)}\otimes h_{(2)}$, $(a\#h)(a'\#h')=\sum a\,(h_{(1)}\cdot a')\#h_{(2)}h'$.	$\tau(h\otimes a)=\sum(h_{(1)}\cdot a)\otimes h_{(2)}$.
Takeuchi smash product $A\#B$	With A an H-module algebra and B an H-comodule algebra, $\rho(b)=\sum b_{(-1)}\otimes b_{(0)}$, $(a\#b)(a'\#b')=\sum a\,(b_{(-1)}\cdot a')\#b_{(0)}b'$.	$\tau(b\otimes a)=\sum(b_{(-1)}\cdot a)\otimes b_{(0)}$.
Ore extension $A[x;\sigma,\delta]$	Multiplication is determined by $xa=\sigma(a)x+\delta(a)$ and multiplication in A.	Set $B=k[x]$: $\tau(x\otimes a)=\sigma(a)\otimes x+\delta(a)\otimes 1$.
Direct-product group algebra $k[N\times Q]$	$(n\otimes q)(n'\otimes q')=nn'\otimes qq'$	$A=k[N],\ B=k[Q]$ $\tau(q\otimes n)=n\otimes q$
Semidirect-product group algebra $k[N\rtimes_\varphi Q]$	$(n\otimes q)(n'\otimes q')=n\varphi_q(n')\otimes qq'$	$A=k[N],\ B=k[Q]$ $\boxed{\tau(q\otimes n)=\varphi_q(n)\otimes q}$

Throughout, let $k=\mathbb{F}_2$ for all the QEC codes.

Table 3: Group-Algebra Realizations of CSS Codes: Trivial Twisting Maps

Name of Code	**Underlying Algebra**	**As a Tensor Product**
HGPS $[[98,18,4]]$	$R=k[C_7\times C_7]$.	$R\cong A\otimes_k B$, where $A=k[x]/(x^7-1)$, $B=k[y]/(y^7-1)$.

Name of Code	**Underlying Algebra**	**As a Tensor Product**
Bivariate bicycle (BB) codes	$R = k[C_\ell \times C_m]$.	$$R \cong A \otimes_k B,$$ where $$A = k[x]/(x^\ell - 1),$$ $$B = k[y]/(y^m - 1).$$
C4 / Iceberg $[[4, 2, 2]]$	$R = k[C_2^2]$. Let $$I = (1 + x, 1 + y).$$ The chain complex is $$0 \longrightarrow I^2 \hookrightarrow R \xrightarrow{\varepsilon} k \longrightarrow 0.$$	$$R \cong A \otimes_k B,$$ where $$A = k[x]/(x^2 - 1),$$ $$B = k[y]/(y^2 - 1).$$
Six-qubit Iceberg $[[6, 4, 2]]$	$R = k[C_6]$.	Since $C_6 \cong C_2 \times C_3$, $$R \cong A \otimes_k B,$$ where $$A = k[x]/(x^2 - 1),$$ $$B = k[y]/(y^3 - 1).$$
Eight-qubit color code $[[8, 3, 2]]$	$R = k[C_2^3]$. Let $$I = (1 + x, 1 + y, 1 + z).$$ The chain complex is $$0 \longrightarrow I^2 \hookrightarrow R \xrightarrow{\varepsilon} k \longrightarrow 0.$$	$$R \cong A \otimes_k B \otimes_k C,$$ where $$A = k[x]/(x^2 - 1),$$ $$B = k[y]/(y^2 - 1),$$ $$C = k[z]/(z^2 - 1).$$
Many-hypercube $[[16, 4, 4]]$	$R = k[C_2^4]$.	$$R \cong A \otimes_k B,$$ where $$A = \frac{k[x, y]}{(x^2 - 1, y^2 - 1)},$$ $$B = \frac{k[s, t]}{(s^2 - 1, t^2 - 1)}.$$

Table 4: Code with a Nontrivial Twisting Map

Name of Code	**Underlying Algebra**	**As a Twisted Tensor Product**
Eight-qubit Gottesman $[[8,3,3]]$	Non-CSS: stabilizers mix X, Y, and Z. For $a, b, a', b' \in \mathbb{F}_2^8$, $(X^a Z^b)(X^{a'} Z^{b'})$ $= (-1)^{b \cdot a'} X^{a+a'} Z^{b+b'}$.	Over $\mathbb{C}$, let $A = \mathbb{C}[C_2^8]$ (X-operators), $B = \mathbb{C}[C_2^8]$ (Z-operators). Then $\boxed{R = A \otimes_\tau B,}$ where $\boxed{\begin{aligned}&\tau : B \otimes A \longrightarrow A \otimes B,\\ &\tau(Z^b \otimes X^a)\\ &\quad = (-1)^{a \cdot b} X^a \otimes Z^b.\end{aligned}}$ This is a genuine nontrivial twisting map satisfying the unit and compatibility conditions. It describes the ambient Pauli operator algebra, not the specific stabilizer code by itself.
Non-Abelian 2BGA $[[12,2,2]]$	Let $A = \dfrac{\mathbb{F}_2[x]}{(x^3-1)}, \qquad B = \dfrac{\mathbb{F}_2[y]}{(y^2-1)}.$	The nontrivial twisting map is $\boxed{\tau(y^j \otimes x^i) = x^{(-1)^j i} \otimes y^j.}$ Thus $yx = x^{-1}y.$ The twist satisfies the unit and compatibility conditions because $x \mapsto x^{-1}$ is an automorphism of order 2. Here $\dim_{\mathbb{F}_2} R = 6$. Both check matrices have rank 5, giving $k = 12 - 10 = 2$ and distance 2.

6 References

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